

Software Engineering

(UE24CS341A)

TEAM NUMBER: 007

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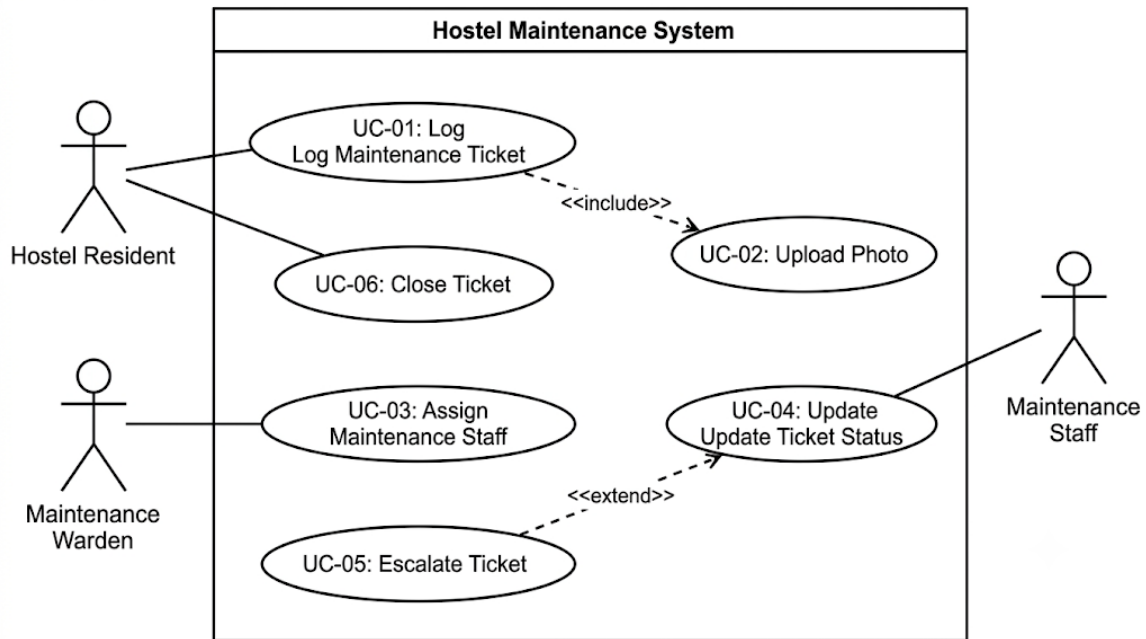
Lab 1: Hostel Maintenance & Issue Ticketing System

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue; Fail: Incomplete form accepted without room tag.	Core functionality for issue reporting
FR-002	Functional	The system shall allow the Maintenance Warden to assign an open ticket to a specific Maintenance Staff member.	High	Pass: Warden can select staff from a dropdown and ticket status changes to Assigned.	Necessary for dispatching staff
FR-003	Functional	The system shall allow Maintenance Staff to update the status of an assigned ticket (e.g., In Progress, Resolved).	High	Pass: Staff updates status to Resolved and timestamp is recorded.	Tracks progress of the issue
FR-004	Functional	The system shall require the Hostel	Medium	Pass: Ticket status changes to Closed	Ensures resident

		Resident to digitally sign off to close a resolved ticket.		after resident confirms.	satisfaction before closure
FR-005	Functional	The system shall display a dashboard of all active tickets for the Maintenance Warden, sortable by severity and time.	Medium	Pass: Warden sees a list of tickets sortable by severity.	Allows warden to oversee and prioritize work
NFR-001	Non-functional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures urgent issues are not ignored
NFR-002	Non-functional	The system shall have an uptime of 99.9% during the academic semester.	Medium	Pass: Uptime logs show ≤ 43 minutes of downtime per month.	Residents must be able to report urgent issues anytime

2. UML Use-Case Diagram



3. Actors & Use Cases

Actors: Hostel Resident, Maintenance Warden, Maintenance Staff

Use-Case Flow Specification

Use Case: UC-01 Log Maintenance Ticket

Primary Actor: Hostel Resident

Preconditions: Resident is authenticated and logged into the system.

Postconditions: A new maintenance ticket is created and placed in the Maintenance Warden's queue.

Main Success Scenario:

1. Resident navigates to the "New Maintenance Ticket" section.
2. System displays the ticket submission form.
3. Resident enters the room number, issue category (plumbing, electrical, carpentry), and severity (low, medium, high).
4. Resident uploads a photo of the issue (Include: UC-02 Upload Photo).
5. Resident submits the ticket form.
6. System validates the inputs and generates a unique Ticket ID.
7. System routes the ticket to the Maintenance Warden's queue.
8. System displays a success message with the Ticket ID to the Resident.

Alternate Flow:

- 4a. Photo Upload Fails (e.g., file too large)
 1. System displays an error message indicating the file exceeds the maximum size.
 2. Resident is prompted to either upload a smaller image or skip the photo attachment.
 3. Resume at step 5.